

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: 7/31/2018 7:31:26 AM
To: "Popovich, Bradley (DEQ)" <PopovichB@michigan.gov>; "Jeff Dietlin" <utilities@Cadillac-MI.net>
Cc: "Brady, Donal (DEQ)" <BradyD6@michigan.gov>
Subject: RE: Cadillac WWTP NPDES Permit Renewal
Attachments: Cadillacs wwtp history 2017.doc, Process Flow Diagram.pdf

Bradley,

Attached you will find an updated flow diagram and general narrative of the treatment plant. If you need anything else please feel free to contact me.

Thanks Doug

Douglas Langworthy

Assistant Water Resources Supervisor

City of Cadillac

231-775-2841

From: Popovich, Bradley (DEQ) [mailto:PopovichB@michigan.gov]
Sent: Thursday, July 19, 2018 1:33 PM
To: Doug Langworthy; Jeff Dietlin
Cc: Brady, Donal (DEQ)
Subject: RE: Cadillac WWTP NPDES Permit Renewal

Good Afternoon Doug and Jeff,

I am currently in the process of reissuing Cadillac WWTP's NPDES Individual permit and have a few things that I'd like to follow up with you on. Please reference the email correspondence below in which Don requests upgraded flow diagrams that include use of UV. Additionally, in order to move further with the permitting process an updated narrative description that includes mention of UV's involvement within your treatment system is needed.

Please provide the updated flow diagram and narrative description to me by **August 2, 2018**. If you have any questions, comments, or concerns please let me know, I'd love to help in any way that I can!

Bradley Popovich

Environmental Quality Analyst

Water Resources Division

Municipal Permits Unit

Michigan Department of Environmental Quality

(517) 284-6656

PopovichB@Michigan.gov

Monday – Friday 8:00-4:30

From: Brady, Donal (DEQ)
Sent: Monday, July 16, 2018 2:54 PM
To: Popovich, Bradley (DEQ) <PopovichB@michigan.gov>
Subject: FW: Cadillac WWTP NPDES Permit Renewal

FYI.



Don Brady, P.E.
Michigan Department of Environmental Quality
Water Resources Division
Cadillac District Office
120 West Chapin Street
Cadillac, MI 49601
BradyD6@michigan.gov
(231) 876-4478

From: Brady, Donal (DEQ)
Sent: Friday, January 5, 2018 6:18 PM
To: Doug Langworthy <dlangworthy@Cadillac-MI.net>; lab@cadillac-mi.net
Cc: Schmitt, Glen (DEQ) <SCHMITTG1@michigan.gov>
Subject: Cadillac WWTP NPDES Permit Renewal

Hi Doug and Cindy,

The DEQ is working on the NPDES permit renewal for the Cadillac WWTP and a couple questions came up.

1. Copper analysis detection limit – As Cindy and I discussed, the current detection limit is 50 ug/L but the plant's lab may be able to achieve a lower limit. The new permit will likely include limits for copper lower than 50 ug/L. Could you please look into this and let me know what detection limit you think you could achieve? If you would like to start conducting the copper analysis with the lower limit, DEQ's permit staff will consider these results.
2. UV Disinfection - The process flow diagrams included with the renewal application showed chlorine disinfection. I know the plant uses UV. Do you have any updated diagrams you could send us?

Thanks for looking into this. Please let me know if you have any questions.



Don Brady, P.E.
Michigan Department of Environmental Quality
Water Resources Division
Cadillac District Office

120 West Chapin Street
Cadillac, MI 49601
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ATTACHMENT NAME:

Cadillacs wwtp history 2017.doc

ATTACHMENT TYPE:

Microsoft Office 1-11, Binary format: document (DOC)

Cadillac's WWTP- a short History

The City of Cadillac WWTP is an advanced treatment plant providing treatment to the City of Cadillac and to several surrounding townships. The daily flow for 2017 averaged 2.01 MGD. The treatment process is comprised of grit removal, primary treatment, activated sludge, secondary treatment, and RBC treatment for ammonia removal, Cloth media filtration and Ultra Violet (UV) Disinfection. The effluent is discharged into the Clam River which is designated as a cold water trout stream several hundred yards from our discharge.

All solids are removed from the primary tanks to the anaerobic digester. From the primary digester solids are transferred to a secondary covered digester which is not heated. This tank is used for solids separation and supernatant withdrawal. Solids from this tank are pumped to one of two storage tanks. Two times a year the city contracts with a sludge application firm for solids injection on farm lands.

Cadillac Has treated sewage for over one hundred years. In 1908 the first plant was built in Cadillac that consisted of five filter beds that used slag for filter media.

In 1923 a new plant was built. This facility utilized a septic tank with a capacity of 444,312 gallons, a 5600 sq. Ft sludge bed and contact filter with 118,272 cu. Ft. of slag filter available. In 1939 5 ½ acres of sand filters operated in series was added.

In 1962 a new “Activated Sludge Treatment Plant” was built. With expansions in 1975, 1989 and 1996 the current plant was established, with upgrades in 2003 and 2007.

The City of Cadillac looks forward to the future challenges that will occur in the wastewater field. Cadillac has always had an interest in providing the best treatment process for its wastewater and has tried to do this for the past 105 years. With any luck Cadillac will be able to continue its stewardship of the public's water for years to come.

Cadillac WWTP

1975 Expansion:

Tertiary building was built
RBC'S were installed
Sand filters were installed

1990-1991 Expansion:

Secondary clarification was added (0.09 MG)
Biosolids storage was added (0.50)
Equalization Basin was installed (0.9 MG)
Main building addition
Influent Screw pump added

1995-1996 Expansion

Primary clarification was added (0.17 MG)
Biosolids storage was added (0.7 MG)
Secondary clarification was added (0.09 MG)
Fine bubble aeration was added (0.33 MG)
Grit removal retrofit
Main building and garage addition

2003 Upgrade:

Replaced chlorine disinfection with U.V. disinfection
Tertiary screw pumps were replaced

2006-2008 Upgrades:

Sand filters replaced with cloth media (Aqua Disk)
Additional U.V. banks installed
Influent screw pumps replaced
Digester retrofitted with a Vaughn Chopper pump
Floating Digester gas holding covers replaced
Biosolids tank #3 was resurfaced
Lake, Leeson and Leisure lift stations were retrofitted with new pumps, controls, generators and
Lake had a new force main installed

ATTACHMENT NAME:

Process Flow Diagram.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image



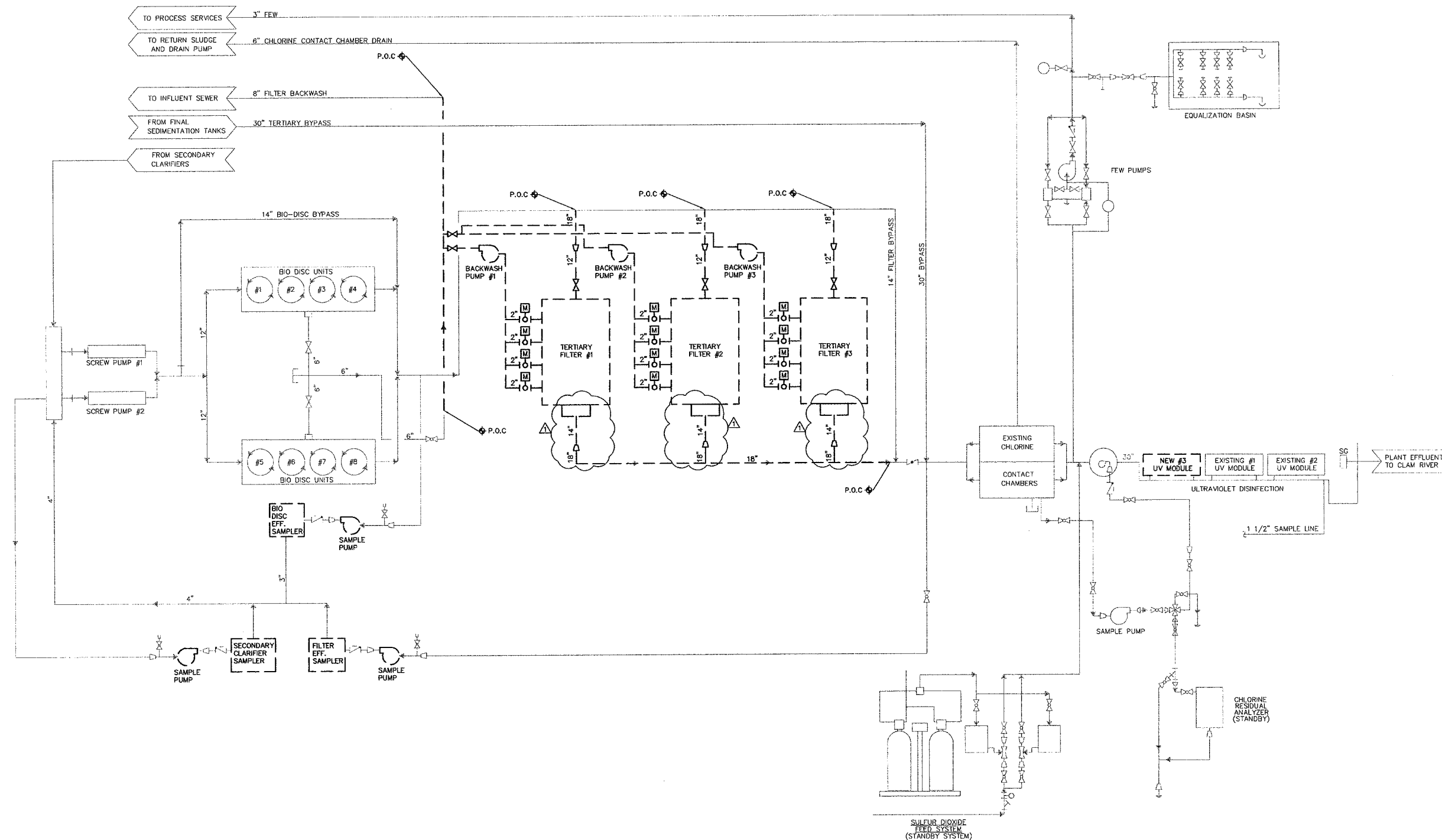
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LEGEND

- NEW
- EXISTING
- ◆ P.O.C. POINT OF CONNECTION

Revision	By	Appd.	YY.MM.DD
CONSTRUCTION SET	WA	IMN	07.11.09
ADDENDUM #1	WA	IMN	07.07.13
BID SET	WA	IMN	07.06.22
FINAL REVIEW SET	WA	IMN	07.05.15
DRAFT REVIEW SET	WA	IMN	07.03.02
Issued	By	Appd.	YY.MM.DD

File Name:	GP1-01 ProcessFlow	JLP	IMN	WA
		Dwn.	Chkd.	Dsgn.

Permit-Seal

Client/Project
CITY OF CADILLAC, MICHIGAN
2007 WASTEWATER TREATMENT PLANT
AND COLLECTION SYSTEM
IMPROVEMENTS
CADILLAC, MI

Title
PROCESS FLOW
DIAGRAM III

Project No.
2075006103

Drawing No.

OP1.03